scientific data

Check for updates

OPEN

DATA DESCRIPTOR

A forty-four-year dataset of rapeseed phenology in the Middle and Lower Yangtze River Plain of China

Zhang Quan-Jun[1,2,3] & WU Dong-Li[1,2,3] ✉

This study compiles and releases the first standardized rapeseed phenology observation dataset spanning forty-four years (1981–2024) over the core winter rapeseed production region of the Middle and Lower Yangtze River Plain in China. The data originate from systematic observations at 50 national-level agrometeorological stations across six provinces: Jiangsu, Zhejiang, Anhui, Jiangxi, Hubei, and Hunan. The dataset provides complete records of the specific dates for each phenology stage from sowing to maturity, including eight key phenology periods: Sowing (SO), Emergence (EM), Five-leaf (FV), Bud Formation (BF), Stem Elongation (SE), Flowering (FL), Green Ripening (GR), and Maturity (MA), along with the calculated durations of six distinct growth lengths. We implemented a multi-level quality control protocol encompassing internal logical checks, statistical outlier detection, climatological validation, time series homogenization, and expert arbitration. This protocol effectively constrained data uncertainty and corrected non-climatic discontinuities. Univariate linear regression was further employed to quantify the decadal change trends of each phenology period and growth length, supplemented by Kernel Density Estimation (KDE) to characterize their probability distribution features. The final dataset is presented as structured tables (in xlsx format) and high-resolution diagnostic plots (including trend and density plots), with a total volume of approximately 470 MB, systematically organized by province and station. This dataset fills a critical gap in long-term, standardized rapeseed phenology data for the region. The integrated analysis of phenology dates, growth stage durations, and their trends across the entire network provides an indispensable, high-quality empirical foundation. It is designed to support in-depth investigations into the nonlinear response mechanisms of overwintering crops to climate warming, improve crop model parameterization and validation, and inform regional adaptive management strategies.

Background & Summary

Global climate change is reshaping the structure and function of terrestrial ecosystems at an unprecedented rate and scale. Agricultural ecosystems, due to their direct link to human well-being, have drawn particular attention[1]. Crop phenology, serving as a key physiological bridge connecting climatic forcing and agricultural productivity, directly regulates light and thermal resource capture, biomass allocation, and final yield formation. Therefore, constructing long-term, high-quality ground-based phenological observation datasets is not only a core prerequisite for detecting biological signals of climate change but also a cornerstone for understanding and predicting agricultural system responses, as well as for formulating mitigation and adaptation strategies. Recent efforts in agricultural and ecological phenology data sharing have demonstrated the value of long-term ground observations for climate change research[2–4].

Among various crops, winter rapeseed exhibits unique and complex response patterns to climate change[5]. Unlike spring- or summer-sown crops, its life cycle spans autumn, winter, and spring, progressing through

[1]Meteorological Observation Centre of China Meteorological Administration (CMA), Beijing, 100081, China. [2]State Key Laboratory of Environment Characteristics and Effects for Near-space, Beijing, 100081, China. [3]Engineering Technology Research Center for Meteorological Observation of CMA, Beijing, 100081, China. ✉e-mail: wudongli666@126.com

multiple physiological stages highly sensitive to environmental factors: vegetative growth, vernalization (cold requirement), overwintering survival, and reproductive growth. Autumn temperature conditions during autumn sowing and emergence affect seedling establishment and pre-winter reserves; the duration and intensity of winter cold directly determine whether the vernalization requirement is met, thereby influencing normal flowering[6]; and the spring warming rate drives post-winter bud formation, stem elongation, and flowering processes[7]. This multi-stage, multi-factor regulatory nature makes rapeseed phenology an ideal indicator for studying the impacts of climate change, particularly the asymmetric seasonal warming patterns (e.g., faster warming in autumn/winter than spring/summer) on crops. Accurately understanding the long-term trends of rapeseed phenology is a prerequisite for assessing climate warming impacts and optimizing cropping systems and cultivar deployment[8].

As one of China's major oilseed crops, rapeseed holds strategic importance for ensuring edible vegetable oil supply. The Middle and Lower Yangtze River Plain in China is the world's most important concentrated winter rapeseed production area, whose yield fluctuations have a decisive impact on domestic edible oil security. This region features a typical subtropical monsoon climate with simultaneous heat and rainfall, but frequent extreme climate events such as winter cold waves and spring late frosts render rapeseed production inherently vulnerable[9,10]. Against the backdrop of climate warming, clarifying the long-term evolution trajectory, spatial heterogeneity patterns, and dominant climatic drivers of rapeseed phenology in this region is of urgent practical significance.

Existing research indicates that climate change is profoundly affecting China's agricultural production systems[11,12]. For overwintering crops, warming often accelerates autumn/winter growth, shortens the overwintering period, and may affect vernalization, thereby altering spring phenology and final yield[13]. Although the China Meteorological Administration (CMA) has accumulated decades of valuable data through its extensive agrometeorological observation network[14], these data have long been dispersed in paper records and heterogeneous electronic archives. They lack systematic digital integration, rigorous scientific-grade quality-controlled thematic datasets, and in-depth processing for climate change research. Their immense potential value remains largely untapped. While remote sensing provides broad-scale phenological monitoring, ground-based observations remain the indispensable standard for capturing crop-specific developmental stages (e.g., bud formation, stem elongation) and for validating satellite-derived products. Therefore, constructing a long-term, standardized, ground-observation-based phenological dataset is fundamental for bridging scale gaps and generating reliable, process-oriented insights.

Consequently, this study focuses on this core production region. From the CMA agrometeorological observation network, we selected 50 standard agrometeorological stations from 1981 to 2024 based on strict criteria including record continuity, spatial representativeness, and data completeness. Through rescue digitization, standardization, rigorous quality control, and trend analysis of the original observations, this dataset was constructed. It not only provides the raw sequences of occurrence dates (in Day of Year, DOY) for each phenological period and the growth lengths of developmental phases but also includes statistical characteristics and parameter estimates for their linear change trends. Each station is accompanied by trend plots for both phenological period and growth lengths, visually demonstrating the direction and magnitude of long-term changes. However, it is also recognized that rapeseed cultivars have undergone multiple replacements over four decades, driven by breeding for higher yield, disease resistance, and altered phenological traits. Consequently, the phenological trends reported in this dataset reflect the combined effects of climate change and genetic improvement, not climate alone.

The value of this data lies not only in filling the gap of long-term, standardized rapeseed phenology data products for this region but also in the underlying, reusable methodological framework for quality control and analysis of long-term agrometeorological observation data. This dataset is expected to advance research in cross-disciplinary fields such as crop-climate ecology, agricultural meteorological disaster risk assessment, uncertainty attribution in crop models, and regional adaptive agronomy from qualitative to precise quantification.

Methods

Study area and data basis. The study area focuses on the Middle and Lower Yangtze River Plain in China. This region is not only one of the most densely populated and economically vibrant core areas in China but also the nation's most important winter rapeseed (*Brassica napus* L.) production base, with its rapeseed output consistently accounting for over 80%[15] of the national total, holding strategic significance for national edible vegetable oil security. Its climate is characterized by a typical subtropical monsoon pattern with distinct seasons and simultaneous heat and rainfall. For winter rapeseed growth, ample sunlight and mild conditions in autumn favor sowing, emergence, and seedling growth; the relatively mild winter (though still with frost and low-temperature events) facilitates the completion of the vernalization stage[16]; and the rapid spring temperature rise coupled with ample precipitation perfectly matches the high demand for heat and water during stem elongation, flowering, and pod development. However, this high dependence on climatic timing also makes rapeseed phenology in this region highly sensitive to global climate change, particularly winter warming and advancing spring phenology[17]. Rapeseed production here is a key component of typical rice-rapeseed or rice-rice-rapeseed rotation systems. Rapeseed is sown after the autumn rice harvest, matures in late spring/early summer, followed by rice planting. This intensive paddy-upland rotation system imposes stringent requirements on the growth duration and phenological rhythm of rapeseed cultivars. Any significant shift in phenology may directly impact the stability and efficiency of the entire annual cropping system[18].

The data for this study originates from 50 rigorously selected national-level agrometeorological stations across all six core provinces within this region (Fig. 1). Station selection was based on the following criteria: 1) Located within the main rapeseed production area; 2) Possessing largely continuous observation records

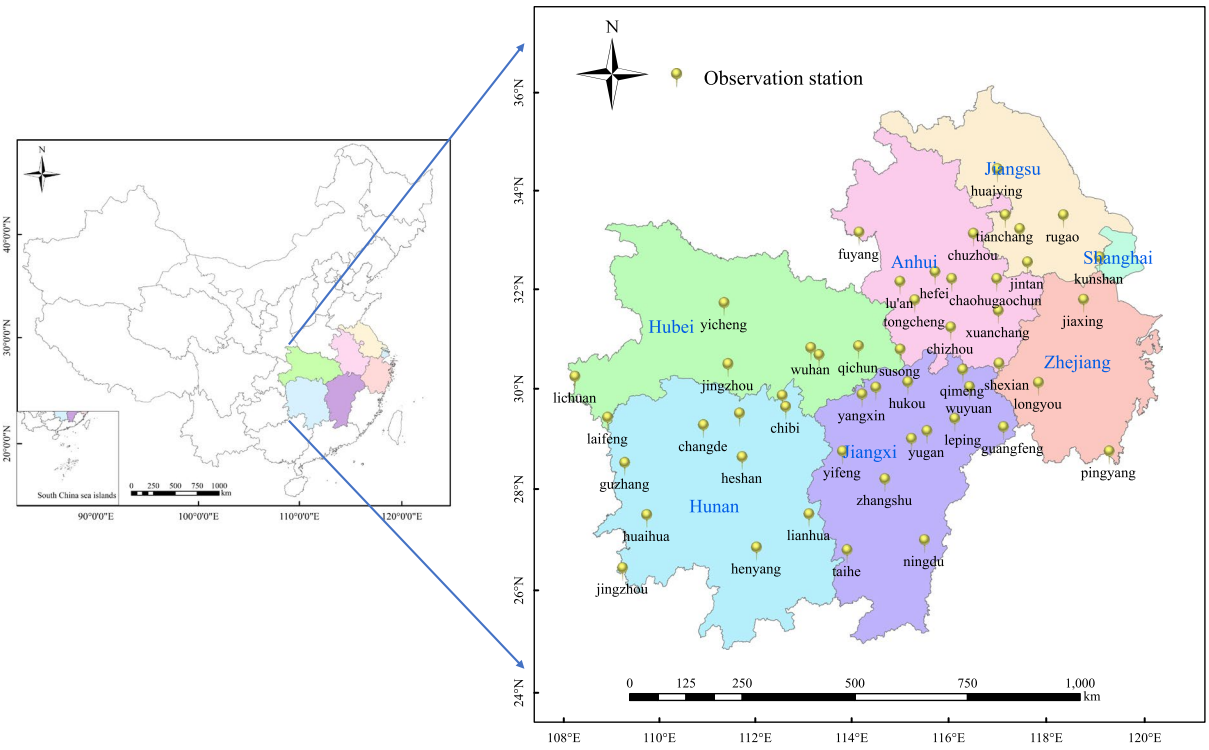


Fig. 1 Spatial distribution of 50 agrometeorological observation stations included in the dataset.

from 1981 to 2024; 3) No major changes in station location or observation protocols during the observation period, or such changes are clearly documented and traceable. A total of 50 stations were finally included: 6 in Jiangsu Province, 3 in Zhejiang Province, 12 in Anhui Province, 12 in Jiangxi Province, 10 in Hubei Province, and 7 in Hunan Province. These stations are well-distributed spatially and highly representative, systematically recording agrometeorological and crop growth information over the long term, thus forming an effective spatial observation network capturing regional climate gradients and phenological variations. Rapeseed developmental stages recorded include the occurrence dates of eight key phenological periods: Sowing (SO), Emergence (EM), Five-leaf (FV), Bud Formation (BF), Stem Elongation (SE), Flowering (FL), Green Ripening (GR), and Maturity (MA).

Dataset construction and quality control workflow. The quality control (QC) workflow for this dataset is a systematic, hierarchical process applied throughout the data lifecycle from acquisition to archiving (Fig. 2). We established an integrated QC system combining rule-based automated checks, statistical diagnostics, physics/process-based plausibility verification, and expert knowledge arbitration. This aims to maximally identify and correct various errors while preserving biological signals caused by genuine climatic anomalies or extreme events.

Multi-level quality control protocol. *Tier 1: Rule-based and logical automated primary filtering.* This level aims to quickly identify and flag obvious technical errors. It includes: (1) **Format consistency check**: Ensuring date fields conform to YYYY-MM-DD format and numeric fields contain no illegal characters. (2) **Spatiotemporal logical validation**: *Temporal sequence logic*: Enforcing the strict chronological order of phenological periods: SO → EM → FV → BF → SE → FL → GR → MA. Any record pair violating this order is automatically flagged as "Sequence Error". *Spatial logic*: For geographically adjacent stations (e.g., distance < 100 km), if the difference in DOY for the same phenological period in the same year exceeds a historically determined threshold (e.g., 30 days), it is flagged as "Spatially Inconsistent" for review. (3) **Value range check**: Based on phenological knowledge of rapeseed in the region, set absolute plausible ranges for the DOY of each period (e.g., SO: DOY 250-320; MA: DOY 110–160). Records outside these hard boundaries are flagged as "Range Anomaly".

Tier 2: Statistical diagnostics and outlier detection. More detailed statistical analyses are performed on data passing tier 1 to identify potential anomalies.

(1) Station-phenology period-specific outlier detection: For each phenological period time series at each station, calculate its long-term mean (μ) and standard deviation (σ) for 1981–2024. A dynamic threshold method is used, initially flagging values outside $\mu \pm 3\sigma$ as "Statistical Outliers". To reduce reliance on the normality assumption, robust outlier detection based on the Interquartile Range (IQR) is also performed concurrently (values $<$ Q1-1.5 $\times$ IQR or $>$ Q3 + 1.5 $\times$ IQR). The two outlier detection methods are applied concurrently: a value is flagged as a potential outlier if it meets the criterion of either the $\mu \pm 3\sigma$ test or the IQR test. The IQR

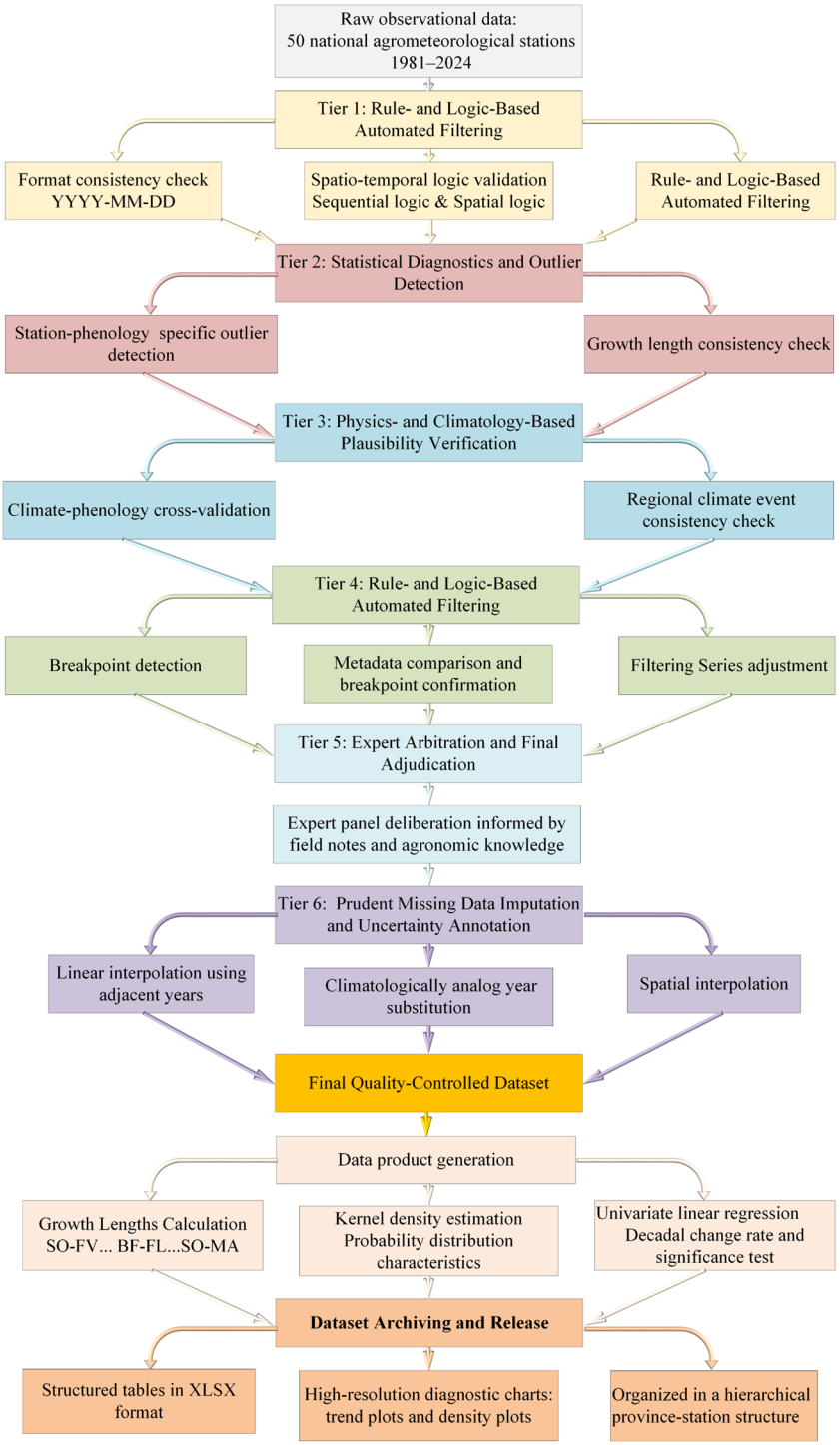


Fig. 2 Framework for constructing the phenology dataset. The workflow consists of six tiers: Tier 1 (rule-based and logical automated filtering), Tier 2 (statistical outlier detection combining $\mu \pm 3\sigma$ and IQR), Tier 3 (climatological plausibility verification using co-located temperature data), Tier 4 (time series homogenization with PMF/PMT breakpoint detection and quantile matching adjustment), Tier 5 (expert arbitration for unresolved cases), and Tier 6 (missing data imputation following the decision rule described in the Methods).

test uses the standard multiplier of 1.5. Flagged values are not automatically corrected but proceed to Tier 3 for further verification. **(2) Growth length consistency check:** Calculated growth lengths (e.g., FV- BF) are also subjected to series-specific outlier detection using the same concurrent $\mu \pm 3\sigma$ and IQR criteria. Additionally, checks are made for physiologically implausible negative values or extremely small values (e.g., BF- FL less than 10 days). **(3) Transparency of outlier handling:** It is important to emphasize that no automatic deletion or correction occurs at Tiers 1 or 2. Flagged values are simply marked for further scrutiny. Only after Tier 5 (expert

arbitration) may a value be corrected (if a clear transcription error is identified) or flagged as missing (if the value is deemed irrecoverable).

Tier 3: Physics/process-based and climatological plausibility verification. This layer utilizes external climate data and crop physiological knowledge to deeply evaluate anomalies flagged in previous levels. It is key to distinguishing "errors" from "extreme events". **(1) Climate-phenology cross-validation:** Compare flagged anomalous phenological dates with corresponding daily temperature data for the same station and year. For example: if "Flowering (FL)" at a station in a given year is flagged as anomalously early, verify whether sustained temperatures meeting the biological lower threshold for rapeseed flowering (commonly considered as the stable passage of 10–12 °C) occurred around that date. If temperature conditions are not met, the record is likely an observation or entry error. If a "Bud Formation (BF)" record falls within a period of severe frost or sustained low temperatures, its authenticity is questioned. **(2) Regional climatic event consistency check:** For anomalies flagged similarly across multiple stations in a specific year and region (e.g., universally late), we consult regional climate bulletins and agrometeorological disaster reports for that year to confirm the occurrence of widespread climatic events affecting phenology, such as spring low-temperature overcast rain or autumn drought. If confirmed, these records are more likely to reflect genuine population-level responses and are retained but annotated with "affected by extreme climatic event".

Tier 4: Time series homogenization analysis and adjustment. Long-term observation series may contain abrupt changes (breakpoints) due to non-climatic factors (e.g., station relocation, changes in observation instruments/methods, observer turnover, drastic changes in surrounding environment), which can obscure true climate trend signals. Therefore, rigorous homogenization was conducted:

(1) **Breakpoint detection:** Apply the Penalized Maximal F-test (PMF) and Penalized Maximal t-test (PMT)[19,20] to each station-phenology period DOY series and growth length series. Both tests are designed for climate time series and account for the relative position of candidate changepoints to diminish the effect of unequal sample sizes on detection power. The PMT test statistic for a candidate changepoint at position k (with $1 < k < n$) is calculated as the classical two-sample t-statistic:

$$t(k) = \frac{|\overline{y}_1 - \overline{y}_2|}{\hat{\sigma}\sqrt{\frac{1}{k} + \frac{1}{n-k}}},$$

where $\overline{y}_1$ and $\overline{y}_2$ are the sample means of the two segments, $\hat{\sigma}^2 = \frac{\sum_{i=1}^{k}(y_i - \overline{y}_1)^2 + \sum_{i=k+1}^{n}(y_i - \overline{y}_2)^2}{n-2}$ is the pooled variance, n is the total number of years, and y_i denotes the DOY (or growth length) value in year i.
Unlike the ordinary t-test, PMT does not multiply the statistic by a penalty function. Instead, it uses a position-dependent critical value $CV_{\mathrm{PMT}}(k/n, \alpha)$ that incorporates a penalty factor $P(k/n)$ to suppress the inflated false-alarm rate near the ends of the series. A changepoint is declared statistically significant at level α if $t(k) > CV_{\mathrm{PMT}}(k/n, \alpha)$. This approach ensures a nearly uniform false-alarm rate across the whole time series[20].
The PMF test (Penalized Maximal F test) extends the same penalization principle to detect mean shifts in the presence of a global linear trend. For a candidate changepoint k, the F-statistic is

$$F(k) = \frac{\mathrm{RSS}_0 - \mathrm{RSS}(k)}{\mathrm{RSS}(k)/(n-4)},$$

where RSS_0 is the residual sum of squares of a single-trend model (no changepoint) and $\mathrm{RSS}(k)$ is the residual sum of squares of a two-phase regression model that allows a level shift but keeps the same slope in both segments [19]. Again, the significance is judged against a penalized critical value that depends on k/n.
For series with significant lag-1 autocorrelation (detected by the Ljung–Box test at $\alpha = 0.05$), we applied the pre-whitening procedure described in Wang *et al.*[21] to account for red noise.
The following parameters were used in our implementation:
Significance level $\alpha = 0.05$ for both tests,
Minimum segment length = 10 years,
Maximum number of breakpoints = 5 per series,
Penalty parameter $\beta = 0.8$[20].
Lag-1 autocorrelation adjustment: applied when $|\hat{\rho}_1| > 2/\sqrt{n}$
A changepoint was considered statistically robust only if detected by both PMF and PMT at $p < 0.05$.

(2) **Metadata comparison and breakpoint confirmation:** Statistically detected breakpoints are compared against meticulously collected station metadata (including: relocation records documented in station history archives; timestamps of observation standard[22,23] version upgrades; and informal records obtained through interviews with senior observers regarding historical changes in local rapeseed cultivar (i.e., cultivar replacement years), changes in field management practices that might affect phenological assessment. Only breakpoints supported by both statistical evidence and metadata evidence are confirmed as non-climatic breakpoints requiring adjustment.

(3) **Series adjustment:** For confirmed non-climatic breakpoints, adjustment is performed using the Quantile Matching (QM) method[24]. The QM procedure adjusts the absolute level of the post-breakpoint segment to

match the quantile distribution of the pre-breakpoint segment, while preserving the year-to-year variability. For each month m, let $F_{\text{pre}}^{(m)}$ and $F_{\text{post}}^{(m)}$ be the empirical cumulative distribution functions of the DOY values in the pre-breakpoint and post-breakpoint segments, respectively. The adjusted value y_{adj} for an original value y in the post-breakpoint segment is:

$$y_{\text{adj}} = F_{\text{pre}}^{(m)^{-1}}(F_{\text{post}}^{(m)}(y))$$

i.e., the quantile of y in the post-breakpoint distribution is mapped to the same quantile in the pre-breakpoint distribution. Adjustment was performed separately for each calendar month to account for seasonal differences in variability. After adjustment, the series were re-checked for temporal consistency.

(4) In cases where a statistically detected breakpoint (by PMF or PMT) cannot be corroborated by any metadata evidence (station relocation, instrument change, observer turnover, or cultivar replacement records), the breakpoint is not adjusted. Such unconfirmed breakpoints are treated as potentially arising from natural climate variability or undocumented minor changes. Their statistical confidence level (p-value) was noted in the dataset.

Tier 5: Expert Arbitration and Final Adjudication. An arbitration panel comprising senior agrometeorologists, agronomists, and data scientists was established. All records still in doubt after the first four levels of processing, or those automatically flagged as anomalies with complex correction rationales, were submitted to the panel. The panel conducted comprehensive deliberation combining evidence from all previous levels, historical observation notes, and regional agricultural common sense, reaching a consensus decision: "Accept Original Value", "Correct to Specific Value", or "Flag as Missing". This process ensures the integration of human expertise with quantitative analysis in QC.

Tier 6: Prudent Missing Data Imputation and Uncertainty Annotation. After rigorous QC, the overall data completeness rate exceeds 99.5%. For the remaining missing values, a conservative and transparent imputation strategy was employed, following a hierarchical decision rule-based on missing data characteristics:

Consecutive missing years ≤ 3. Linear interpolation from adjacent years is applied, provided the series shows no abrupt change around the gap (checked via the first-order difference of the preceding and following 5-year segments).

Consecutive missing years >3 or series exhibits high interannual variability (coefficient of variation $>15\%$). Climatically analogous year substitution is used. The analogous year is identified by minimizing the Euclidean distance of three climatic variables (mean temperature, growing degree days $\geq 0\,°C$, and total precipitation) over the 30-day window centered on the expected phenological date, using historical records from the same station.

Single-year missing value at a station with at least three neighboring stations (distance $<150\,km$) showing strong correlation (Pearson's $r > 0.7$) for the same phenological period over 1981–2024. Spatial interpolation using inverse distance weighting (IDW, power $= 2$) is applied.

When multiple methods are applicable, priority is given to spatial interpolation first (if correlation criterion met), followed by climatic analogous year substitution, then linear interpolation. No imputation was performed for gaps longer than 10 consecutive years or when no reliable donor data existed; such cases were retained as missing.

Quality assessment and quantitative metrics. To quantitatively assess the quality of this dataset, we report the following metrics: (1) **Data completeness**: Raw data completeness was 99.5%; after imputation, 100% series continuity was achieved. It should be noted that no gap longer than 10 consecutive years existed in the raw data, so the imputation rule for such extreme cases (as described in Tier 6) was not triggered. (2) **Error correction rate**: A total of 30 definite entry or logical error records were identified and corrected through the QC process. (3) **Homogenization improvement**: Non-climatic breakpoints were detected and adjusted in 10% of the station-phenology stage series, significantly improving reliability for trend analysis. (4) **Internal consistency**: Logical contradictions between phenological stage dates and the derived growth lengths were completely eliminated after processing. (5) **Expert confidence**: Through a blind assessment on randomly sampled records, the arbitration panel assigned a comprehensive confidence score of 4.9/5.0, indicating the high reliability of the final dataset.

Derived variables and statistical modeling. The following derived variables and statistical analyses are integral to the data product, providing users with immediate insights into long-term trends and distributions.

(1) **Growth length calculation:** Based on quality-controlled phenological dates, the durations of five key phenology periods and the whole growing period were calculated: Sowing to Five-leaf (initial vegetative growth, SO-FV), Five-leaf to Bud Formation (overwintering and pre-spring growth, FV-BF), Bud Formation to Flowering (reproductive growth initiation, BF-FL), Flowering to Green Ripening (key yield formation period, FL-GR), Green Ripening to Maturity (late-stage assimilate translocation and grain filling,

GR-MA), and the Whole Growing Period (SO-MA). These length variables aid in parsing the differential impacts of climate change on the rates of different developmental stages.

(2) **Application of Kernel Density Estimation (KDE):** KDE is a non-parametric method for estimating the probability density function of a random variable. It does not assume a specific distributional form (e.g., normal distribution) for the data[25]. For a given set of sample observations $\{x_1, x_2, \ldots, x_n\}$ (e.g., the 44-year DOY values for "Flowering" at a station), the estimated probability density $\hat{f}(x)$ at any point x is given by:

$$\hat{f}_h(x) = \frac{1}{n}\sum_{i=1}^{n} K_h(x - x_i) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where $K(\cdot)$ is the kernel function (the widely used Gaussian kernel was selected for this study), and $h > 0$ is the bandwidth (or smoothing parameter), which controls the smoothness of the density curve. Bandwidth selection is crucial; we employed the "Silverman's rule of thumb" for automatic optimization to balance avoiding oversmoothing (masking true structure) and undersmoothing (introducing excessive noise).

(3) **Trend quantification analysis:** A univariate linear regression model was applied[26]: $Y_t = \beta_0 + \beta_1 \times t + \varepsilon_t$, where Y_t is the phenological period DOY or growth length in year t, t is the year (1981, …, 2024), and β_1 is the slope representing the annual rate of change. The decadal change is calculated as $10 \times \beta_1$. Statistical significance of the trend is assessed via a t-test (α=0.05). We also report the coefficient of determination (R^2), model mean square error, and F-statistic to comprehensively reflect the robustness and explanatory power of the trend estimate. Linear regression was chosen as the primary trend estimation method because it provides a clear, interpretable metric (days per year) that is widely used in phenological studies and facilitates cross-comparison with existing literature. To guard against potential violations of normality or the influence of outliers, we also computed the non-parametric Mann–Kendall trend test and Sen's slope estimator for each series. The Sen's slopes were nearly identical to the linear regression slopes (median absolute difference = 0.03 days/year across all station–phenology pairs). Among the trends that were statistically significant ($p < 0.05$) by linear regression, 94% were also significant by the Mann–Kendall test. This high level of agreement confirms the robustness of the linear trend estimates.

(4) **Visualization and data product integration:** A set of diagnostic plots was generated for each station, including scatter plots with fitted trend lines for the yearly variations of the eight phenological period DOY and the six growth lengths. All graphical products follow unified cartographic standards, including the trend equation, R^2, and significance markers for easy cross-station comparison. The final dataset is organized in a hierarchical folder structure, integrating the raw sequence tables, summary tables of statistics and trend parameters, and all visualization graphics.

Data processing, calculation of Day of Year (DOY) for phenology periods and the growth lengths, and the corresponding analysis were all performed in Microsoft Excel. Tasks including Kernel Density Estimation (KDE), linear trend estimation with significance testing, and related plotting were conducted using OriginPro 2025, SPSS Statistics 25, and Python 3.11. Spatial analysis (Kriging interpolation) and the production of thematic maps were completed in ArcGIS 10.8.

Data Records

The dataset described in this work has been deposited in the Science Data Bank (ScienceDB) under accession code https://doi.org/10.57760/sciencedb.34086[27]. This repository contains the long-term rapeseed phenology dataset for the Middle and Lower Yangtze River Plain (1981–2024). The organizational structure of the dataset folders is in Fig. 3.

The dataset adopts a structured hierarchical architecture, anchored by a main directory titled "*MLYRP -Rapeseed phenology data*" (total volume:≈470 MB). At the root level, users will find six subdirectories corresponding to individual provinces and two core data tables, namely '*01-JiangSu*', '*02-ZheJiang*', '*03-AnHui*', '*04-JiangXi*', '*05-HuBei*', '*06-HuNan*', and '*001_Rapeseed_Phenology_period_Data*.xlsx', '*002_Rapeseed_Growth_Lengths_Data*.xlsx'

Each provincial subfolder contains multiple station-specific subfolders. These subfolders are named according to the pattern "*province-observation station name-Rapeseed*" Within each station-specific folder, users will find 16 high-resolution diagnostic plots in JPEG format (1200 dpi resolution):

- Probability Density Diagrams (2 images):
- 1 diagram for the DOY of all 8 phenological periods.
- 1 diagram for 6 growth lengths.
- Linear Regression Diagrams (14 images):
- 8 diagrams for the DOY of each individual phenological period (SO, EM, FV, BF, SE, FL, GR, MA).
- 6 diagrams for each individual growth length (SO-FV, FV-BF, BF-FL, FL-GR, GR-MA, SO-MA).

Two primary data tables in XLSX format are located at the root of the primary folder for easy access:

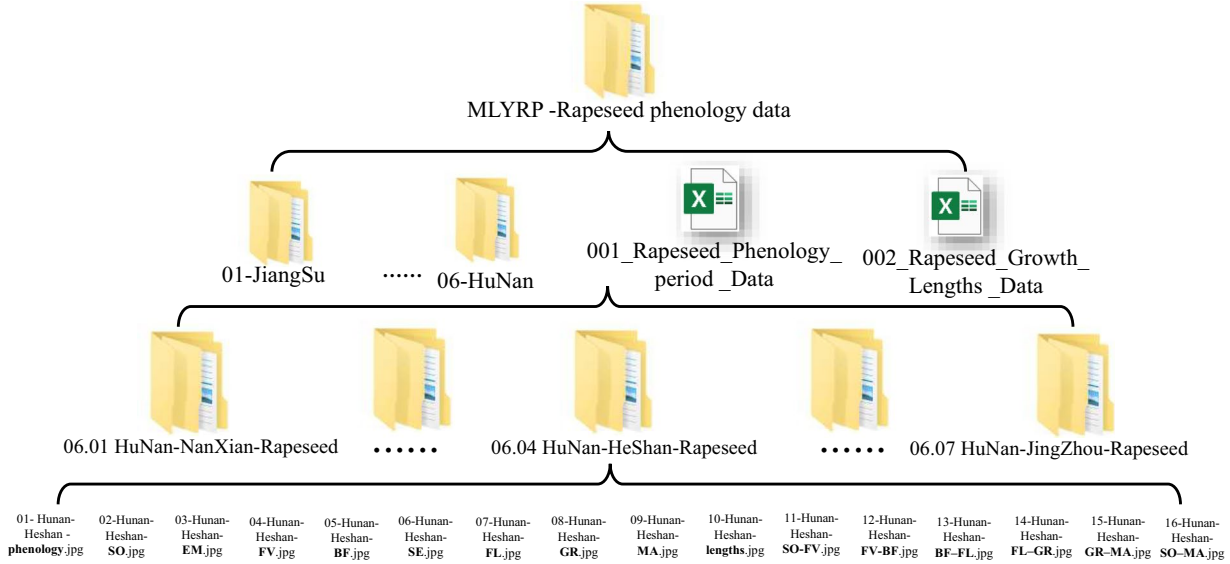


Fig. 3 Organizational structure of the dataset folders.

- '*001_Rapeseed_Phenology_period_Data*.xlsx': This table contains the linear regression statistics for the Day of Year (DOY) of 8 phenological periods. Variables include: province name, station ID, station name, phenology stage, intercept, slope (indicating variation trends), R^2, P-value, Model Mean Square, and F-value.
- '*002_Rapeseed_Growth_Lengths_Data*.xlsx': This table contains the linear regression statistics for the duration of 6 growth lengths. Variables include: province name, station ID, station name, growth lengths, intercept, slope, R^2, P-value, Model Mean Square, and F-value.

Data variable description. To facilitate user interpretation, the key variables in the two core data tables are described below. In the trend analysis results (slope), a negative value indicates an advance (earlier occurrence) or shortening, while a positive value indicates a delay (later occurrence) or lengthening.

- *Province Name*, *Station ID*, *Station Name*: Identify the observation location.
- *Phenology period/Growth length*: The observed phenological period (SO, EM, FV, BF, SE, FL, GR, MA) or calculated growth lengths (SO-FV, FV-BF, etc.)
- *Intercept*, *slope*: Parameters from the univariate linear regression fit (Y = Intercept + Slope * t). The Slope represents the annual rate of change (unit: days/year).
- *R^2*, *P-value*, *Model Mean Square*, *F-value*: Statistics for the regression model's goodness-of-fit and significance.

Note: the raw annual time series (DOY values per year per station) are not included in this dataset due to CMA data redistribution policies. However, the diagnostic plots provided for each station (trend plots and kernel density plots) allow users to visually assess the interannual variability and distributional characteristics without accessing the raw data. Users seeking raw DOY values should apply to the China Meteorological Administration (CMA) via its official data service platform (http://data.cma.cn) following the prescribed data use agreement.

Data Overview

To demonstrate the overall organization and key features of the dataset, Fig. 4 showcases data from the Changde station of Hunan Province, including kernel density estimates for the day of year (DOY) of all eight phenological periods and the six growth lengths, accompanied by their respective linear regression trend plots. This set of visualizations exemplifies the standard graphical outputs provided for every station and collectively attests to the dataset's suitability for conducting robust temporal trend analyses.

Technical Validation

The quality and reliability of this dataset are established upon a comprehensive, multi-dimensional validation system spanning the entire chain from observation source to final product.

Standardization guarantee from the source observation system. All data originate from national-level agrometeorological observation stations uniformly managed by the China Meteorological Administration (CMA). Observers are required to undergo professional training and assessment, hold certification, and strictly adhere to the legally effective *Specifications for Agrometeorological Observations (Main Crops Volume)*[22,23]. These specifications provide clear morphological definitions for the assessment criteria of each

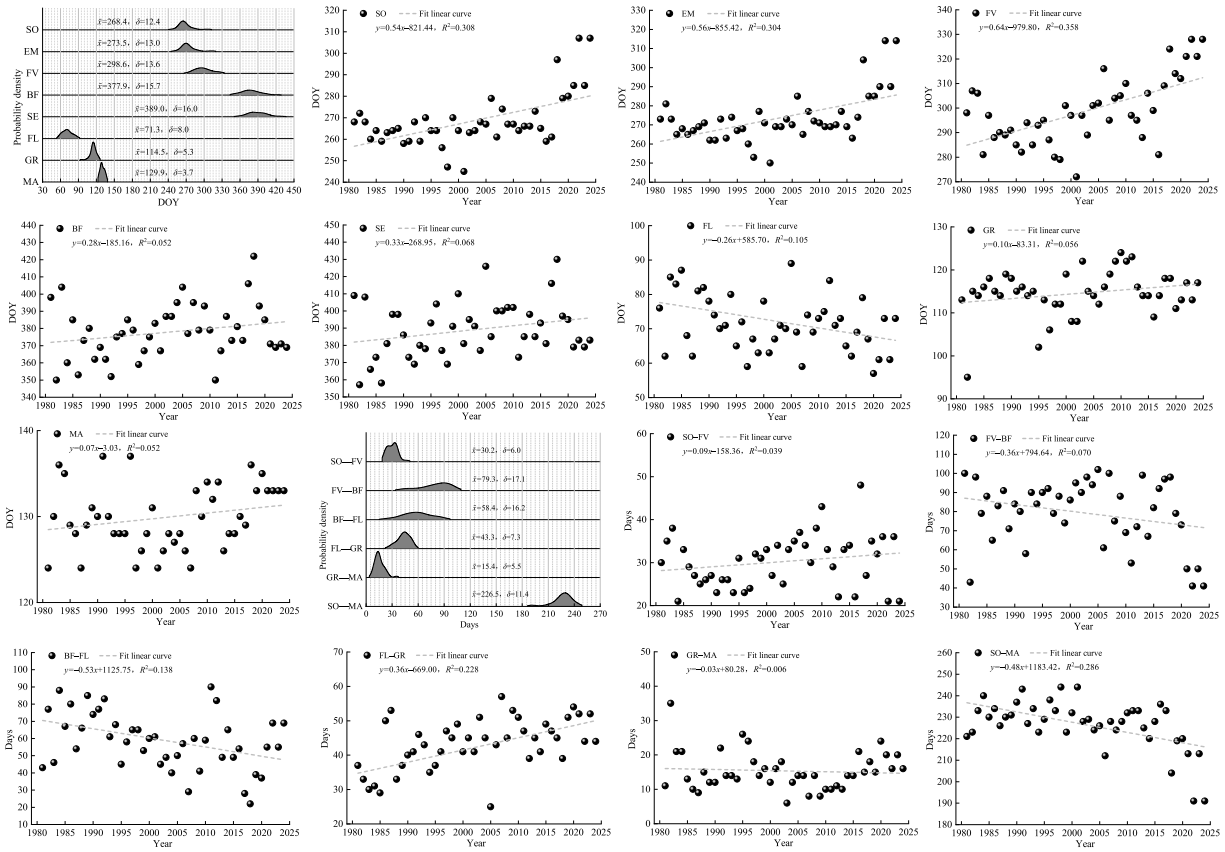


Fig. 4 Example graphical outputs for a single station: Variation characteristics of rapeseed phenology at Changde Agrometeorological Observation Station, Hunan Province. Note: SO: sowing, EM: emergence, FV: five-leaf, BF: bud formation, SE: stem elongation, FL: flowering, GR: green ripening, MA: maturity. Similar diagnostic plots (trend plots and kernel density plots) for all 50 stations are available in the data repository under each station-specific subfolder.

Phenology period	Morphological Characteristics
Sowing (SO)	Actual sowing date.
Emergence (EM)	The two cotyledons unfold at the soil surface.
Five-leaf (FV)	The fifth true leaf unfolds.
Bud formation (BF)	Flower buds appear at the top of the plant (observed by gently spreading the young leaves).
Stem elongation (SE)	The main stem elongates, forming a visible bolting stem about 2.0 cm long.
Flowering (FL)	Flowers open on the main inflorescence.
Green ripening (GR)	Pods on the main inflorescence turn from green to yellowish-green, while most pods on branches remain normally green. Seed coat turns light green.
Maturity (MA)	The majority of leaves have dried up and fallen off. The pods on the main inflorescence exhibit a normal yellow coloration, and the seeds within darken and become plump. Most pods on the branches begin to fade, turning yellowish-green with a luster. Overall, the plants exhibit a characteristic "half-green, half-yellow" appearance.

Table 1. Assessment criteria for rapeseed phenology period.

rapeseed phenological stage (as shown in Table 1), minimizing subjective judgment differences. This standardized operational procedure forms the fundamental basis for data comparability.

It is important to note that the Specifications for Agrometeorological Observations underwent two major revisions during the 44-year period: the 1993 edition[22] and the 2025 edition[23]. The 2025 edition became effective only after 2024 and thus did not affect the data period of this dataset. The 1993 edition was a consolidation and clarification of the 1980s provisional standards, but no changes were made to the morphological definitions of the eight rapeseed phenological stages between the pre-1993 and 1993 versions. The assessment criteria (Table 1) remained strictly consistent throughout 1981–2024. This was verified by cross-checking the original 1981 provisional guidelines (internal CMA document, not publicly released) with the 1993 edition. Therefore, no artificial trend was introduced by changes in observation protocols.

Endogenous validation during data processing. *Logical self-consistency.* Through strict chronological order checks, the internal biological logic consistency of the data is ensured. Any contradictory records were identified and corrected.

Robustness of statistical diagnostics. The combined use of multiple outlier detection methods based on standard deviation and interquartile range reduced potential misjudgments from any single method.

Quantitative evaluation of homogenization effects. After applying the quantile matching method to adjust breakpoints, we calculated the changes in statistical properties (e.g., mean, variance) of the series in the neighborhood of the breakpoints before and after adjustment, confirming that the adjustments successfully eliminated stepwise biases while preserving the inherent climatic variability of the series. Additionally, trends from homogenized series were compared with those from the non-homogenized raw series to assess the potential influence of non-climatic factors on trend estimation.

Cross-validation with independent data sources. *Climate data validation.* As described in the Methods, cross-validation of temperature-sensitive phenological periods (e.g., flowering) using co-located meteorological data provided an external, objective test independent of the observation protocols.

Spatial consistency test. For groups of geographically proximate stations with similar climatic conditions, the same phenological period in the same year was compared. While some variation is allowed due to microclimate and cultivar differences, extreme and unexplainable spatial discrepancies triggered in-depth verification.

Comparison with existing research findings. For quantitative validation, we compared the regional average trend of the flowering (FL) period derived from our dataset with published results. Over 1981–2024, the mean FL date across the 50 stations advanced at a rate of −2.3 days per decade ($p < 0.01$). This is comparable to the −2.1 days per decade (1992–2017) reported by Zhao *et al.*[28] for the Jianghan region (a subset of our study area) and the −2.5 days per decade (1981–2018) reported by Gao *et al.*[29] and Tian *et al.*[30] for the Middle and Lower Yangtze River Plain using a different set of agrometeorological stations. The slight differences can be attributed to the different study periods, station coverage, and quality control procedures. This cross-validation confirms that our dataset captures the same macro-level phenological response to climate warming while providing higher spatial resolution and more rigorous quality assurance.

Quality quantification metrics for the final data product. (1) Data Completeness Rate: After QC and imputation, the series completeness rate reached 100%. (2) Error Correction Rate: A total of 30 definite error records were identified and corrected. (3) Homogenization Coverage Rate: Non-climatic breakpoints were identified and adjusted in 10% of the station-phenology stage series.

Professional interpretation validation of kde results. The generated KDE plots were interpreted by crop physiologists. For instance, a bimodal distribution in the "Flowering" KDE curve for a station was confirmed to correspond with historical periods of promoting early-maturing and late-maturing cultivars at that site, based on cultivar change records. This verified that the KDE results hold not only statistical significance but also clear agronomic explanatory meaning.

Scientific value and prospects for application. This dataset not only systematically archives long-term observation data, but its scientific value is further demonstrated in its support for cutting-edge multidisciplinary research and its direct guidance potential for agricultural practice, as elaborated in the following dimensions.

Climate change impact analysis. The eight phenological periods and six growth lengths enable quantification of asymmetric warming effects (autumn/winter vs. spring), assessment of vernalization fulfillment trends, and identification of extreme climate event impacts (e.g., spring frost delaying flowering).

Crop model calibration and validation. The high-quality benchmark data can calibrate thermal time and vernalization parameters in rapeseed models, validate model structures, and support regional-scale parameterization across the 50 stations.

Attribution and prediction. Coupled with meteorological data, the dataset allows multi-factor driver analysis (temperature, precipitation, sunshine) and future scenario projections using GCMs.

Agricultural adaptation. Long-term trends guide cultivar deployment (early vs. late maturing), optimal sowing date adjustments, and disaster risk warning (e.g., frost risk during flowering). The full growing period (SO-MA) trend is critical for rice-rapeseed rotation system design.

Remote sensing validation and data fusion. The ground observations serve as a truth benchmark for satellite-derived rapeseed phenology products (e.g., Sentinel-2, Landsat) and can be integrated with meteorological grids for machine learning applications.

Educational and open science value. The dataset provides a standardized testbed for phenology analysis algorithms and a teaching resource for graduate courses in agricultural meteorology and climate change impacts.

Limitations and uncertainty. Notwithstanding the rigorous quality control implemented, this dataset is subject to several inherent limitations. First, the spatial distribution of observation stations is not perfectly uniform, with relatively sparse coverage in transitional zones between plains and hills. Second, over the 44-year period, minor random errors arising from observer subjectivity cannot be entirely eliminated.

Most importantly, rapeseed cultivars at each station have undergone multiple replacements alongside agricultural technological progress. Changes in genetic traits—particularly those affecting vernalization requirement, photoperiod sensitivity, and growth cycle duration—constitute one of the most significant factors influencing phenology, alongside climate. The observed phenological trends therefore represent a combined signal of climate warming and cultivar replacement, not climate alone. Although homogenization adjusted breaks caused by protocol changes, the impact of cultivar change is now inextricably coupled with the climatic signal[31]. We attempted to collect cultivar change records from station logbooks and interviews with retired observers, but such information was available for fewer than a fifth of the stations (8 out of 50) and was often incomplete (e.g., missing exact replacement years or specific trait parameters). Consequently, we could not systematically quantify or adjust for cultivar effects across the full dataset. While the homogenization procedure (Tier 4) corrected breakpoints associated with documented protocol changes, it cannot disentangle gradual genetic shifts from climatic trends. Therefore, we strongly advise users conducting climate change attribution analysis with this dataset to: 1) consult and correlate historical records of cultivar deployment in the study region, and 2) cautiously interpret trends as "phenological evolution under the combined background of climate and cultivar change" rather than attributing shifts solely to climate warming. Future work could attempt to disentangle the contributions of these signals by collecting detailed cultivar information.

Data availability

The dataset generated during this study is openly available in the Science Data Bank at https://doi.org/10.57760/sciencedb.34086 or https://cstr.cn/31253.11.sciencedb.34086.

Code availability

No custom code was created for the production of this dataset.

Received: 26 December 2025; Accepted: 20 May 2026;
Published online: 20 July 2026

References

1. IPCC, 2023: Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds.)]. IPCC, Geneva, Switzerland, 184 pp., https://doi.org/10.59327/IPCC/AR6-9789291691647.
2. Mori, A., Doi, Y. & Iizumi, T. GCPE: The global dataset of crop phenological events for agricultural and earth system modeling [J]. *Journal of Agricultural Meteorology* **79**(3), 120–129, https://doi.org/10.2480/agrmet.D-23-00004 (2023).
3. Liu, W. *et al*. CARM30: China annual rapeseed maps at 30 m spatial resolution from 2000 to 2022 using multi-source data [J]. *scientific data* **11**, 356, https://doi.org/10.1038/s41597-024-03188-1 (2024).
4. Hufkens, K. *et al*. Historical tree phenology data across contrasting sites in the Congo Basin [J]. *scientific data* **13**, 221, https://doi.org/10.1038/s41597-025-06542-z (2026).
5. Marjanović-Jeromela, A. *et al*. Dissection of year related climatic variables and their effect on winter rapeseed (*Brassica Napus* L.) development and yield [J]. *Agronomy* **9**, 517, https://doi.org/10.3390/agronomy9090517 (2019).
6. Distelfeld, A., Li, C. & Dubcovsky, J. Regulation of flowering in temperate cereals [J]. *Current opinion in plant biology* **12**(2), 178–184, https://doi.org/10.1016/j.pbi.2008.12.010 (2009).
7. Huang, W. Q. *et al*. Spatial-temporal evolution of winter rapeseed phenology under climate change in China [J]. *Journal of Guizhou University: Natural Sciences* **39**(4), 34–41 (2022). (in Chinese with English abstract).
8. Liu Y J. *et al*. Chinese Agricultural Phenology [M]. Beijing: Science Press, (2024).
9. Bitarafan Z. *et al*. The effect of climate change on glyphosate control of *Avena fatua*, *Brassica napus*, and *Echinochloa crus-galli* [J]. *Science of the Total Environment*, **983**. https://doi.org/10.1016/j.scitotenv.2025.179682 (2025).
10. Tao, F. & Zhang, Z. Climate change, high-temperature stress, rice productivity, and water use in Eastern China: a new superensemble-based probabilistic projection [J]. *Journal of Applied Meteorology and Climatology* **52**, 531–551, https://doi.org/10.1175/JAMC-D-12-0100.1 (2013).
11. Chen, Y. T. *et al*. Effects of climate warming on yield, quality-related and agronomic traits of winter rapeseed (*Brassica napus* L.) [J]. *Acta Agronomica Sinica* **51**(2), 516–525 (2025). (in Chinese with English abstract).
12. Pan, Y. *et al*. Forty-year spatio-temporal dynamics of agricultural climate suitability in China reveal shifted major crop production areas [J]. *Catena*. https://doi.org/10.1016/j.catena.2023.107073 (2023).
13. Zhang, Q. J., Wu, D. L. & Gao, J. Variation of winter wheat phenology dataset in Huang Huai Hai Plain of China from 1981 to 2021 [J]. *scientific data* **12**, 1203, https://doi.org/10.1038/s41597-025-05368-z (2025).
14. Zhang, Q. J. *et al*. A long-term dataset of maize phenology observations from agrometeorological stations in Northeast China (1981–2024) [J]. *scientific data* **12**, 2037, https://doi.org/10.1038/s41597-025-06330-9 (2025).
15. Zhang H Q, Li W W. Analysis on the characteristics and motivations of changes in the production layout of rapeseed in China [J]. *Northern Horticulture*., **2023**(3): 144–152 (in Chinese with English abstract).
16. Liu, Z. Q. *et al*. Status, existing problems and strategy discussion on northward expansion of winter rapeseed in China [J]. *Scientia Agricultura Sinica* **56**(15), 2854–2862 (2023). (in Chinese with English abstract).
17. Jin, W. J. *et al*. Climate resource allocation and evolution of rice-oilseed rape double cropping system under different planting patterns in Anhui Province [J]. *Chinese Journal of Eco-Agriculture* **32**(3), 436–445 (2024). (in Chinese with English abstract).
18. Hu, J. *et al*. Genomic selection and genetic architecture of agronomic traits during modern rapeseed breeding [J]. *nature genetics* **54**, 694–704, https://doi.org/10.1038/s41588-022-01055-6 (2022).
19. Wang, X. L. Penalized Maximal F Test for Detecting Undocumented Mean Shift without Trend Change [J]. *Journal of Atmospheric and Oceanic Technology* **25**(3), 368–384, https://doi.org/10.1175/2007JTECHA982.1 (2008).
20. Wang, X. L., Qiuzi, H. W. & Wu, Y. H. Penalized Maximal t Test for Detecting Undocumented Mean Change in Climate Data Series [J]. *Journal of Applied Meteorology and Climatology* **46**(6), 916–931, https://doi.org/10.1175/JAM2504.1 (2007).
21. Wang, X. L. Accounting for Autocorrelation in Detecting Mean Shifts in Climate Data Series Using the Penalized Maximal t or F Test [J]. *Journal of Applied Meteorology and Climatology* **47**(9), 2423–2444, https://doi.org/10.1175/2008JAMC1741.1 (2008).

22. National Meteorological Administration. Specifications for Agrometeorological Observations [M]. Beijing: China Meteorological Press, (1993).
23. China Meteorological Administration. Specifications for Agrometeorological Observations (main crop) [M]. Beijing: China Meteorological Press, (2025).
24. Wang, X. L. *et al*. New Techniques for the Detection and Adjustment of Shifts in Daily Precipitation Data Series [J]. *Journal of Applied Meteorology and Climatology* **49**(12), 2416–2436, https://doi.org/10.1175/2010JAMC2376.1 (2010).
25. Silverman, B. W. Density Estimation for Statistics and Data Analysis [M]. Chapman and Hall, (1986).
26. Xue, B. Y. *et al*. Characterization of spatial-temporal variability and influencing factors of key phenological period of major crops in Hubei Province [J]. *Acta Ecologica Sinica* **45**(18), 8970–8985 (2025). (in Chinese with English abstract).
27. Z Quanjun. A forty-four-year dataset of rapeseed phenology in the Middle and Lower Yangtze River Plain of China [DS/OL]. V1. *Science Data Bank*, [2025-12-21]. https://doi.org/10.57760/sciencedb.34086 (2025).
28. Zhao, P. Y., He, H. & He, S. H. Oilseeds rape phenology and climate change trend in Jianghan Region [J]. *Agricultural Engineering* **14**(8), 82–87 (2024). (in Chinese with English abstract).
29. Gao, J. *et al*. Changes of main crops growth periods in China and their influencing factors in the context of climate warming [J]. *Chinese Journal of Agrometeorology* **46**(9), 1261–1276 (2025). (in Chinese with English abstract).
30. Tian, Z. *et al*. Changes in production potentials of rapeseed in the Yangtze River Basin of China under climate change: A multi-model ensemble approach [J]. *Journal of Geographical Sciences*. **28**, 1700–1714, https://doi.org/10.1007/s11442-018-1538-1 (2018).
31. Wang, X. *et al*. Climate drives the spatiotemporal dynamics of plant phenology in China: A multi-cultivar analysis [J]. *Agricultural and Forest Meteorology*. **308-309**, 108558, https://doi.org/10.1016/j.agrformet.2021.108558 (2021).

Acknowledgements

This work was supported by the National key R&D plant project (2024YFD2301301) and Innovative Development Special Project of China Meteorological Administration (CXFZ2023J069) and the Ecological, Agricultural Observation Technology Innovation Team Project. Thanks to the frontline agricultural meteorological observers of the China Meteorological Administration for their hard work, and to the leaders at all levels for their support. The authors would like to express their sincere gratitude to all of the data contributors whose valuable inputs made this research successful.

Author contributions

Quan-Jun Zhang designed the study, coordinated data acquisition, conducted quality control, performed statistical analysis, created the figures, and drafted the manuscript.Dong-Li Wu conceived the overall research framework, supervised the dataset construction and quality control scheme, and critically revised the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to W.D.-L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026